

Practical Laboratory 1

Two VM Network Commissioning and Connectivity Verification

WADF105 Network Security Fundamentals

In this laboratory we will build a small, protected network using only the Ubuntu Client machine named icdfa-nslab-client-v1 and the OPNsense Firewall machine named icdfa-nslab-firewall-v1. We will configure the VirtualBox adapters, confirm addressing and routing, reach the firewall management interface, test internet access, and observe ARP, ICMP and DNS traffic in Wireshark.

Learning objectives

- Explain the purpose of WAN and LAN interfaces on a firewall.
- Configure a two-VM VirtualBox network without relying on fixed MAC addresses.
- Obtain or assign a valid IPv4 address, default gateway and DNS server.
- Verify local connectivity, routing, name resolution and internet access.
- Capture and identify ARP, ICMP and DNS packets in Wireshark.

Target topology

Component	VirtualBox connection	Expected addressing	Purpose
icdfa-nslab-firewall-v1 WAN	Adapter 1: NAT	DHCP from VirtualBox, commonly 10.0.2.15/24	Internet-facing laboratory interface
icdfa-nslab-firewall-v1 LAN	Adapter 2: Internal Network named ICDFA-LAN	10.10.10.1/24	Protected network and default gateway
icdfa-nslab-client-v1	Adapter 1: Internal Network named ICDFA-LAN	DHCP, normally 10.10.10.100-200/24	Student workstation and packet-analysis host

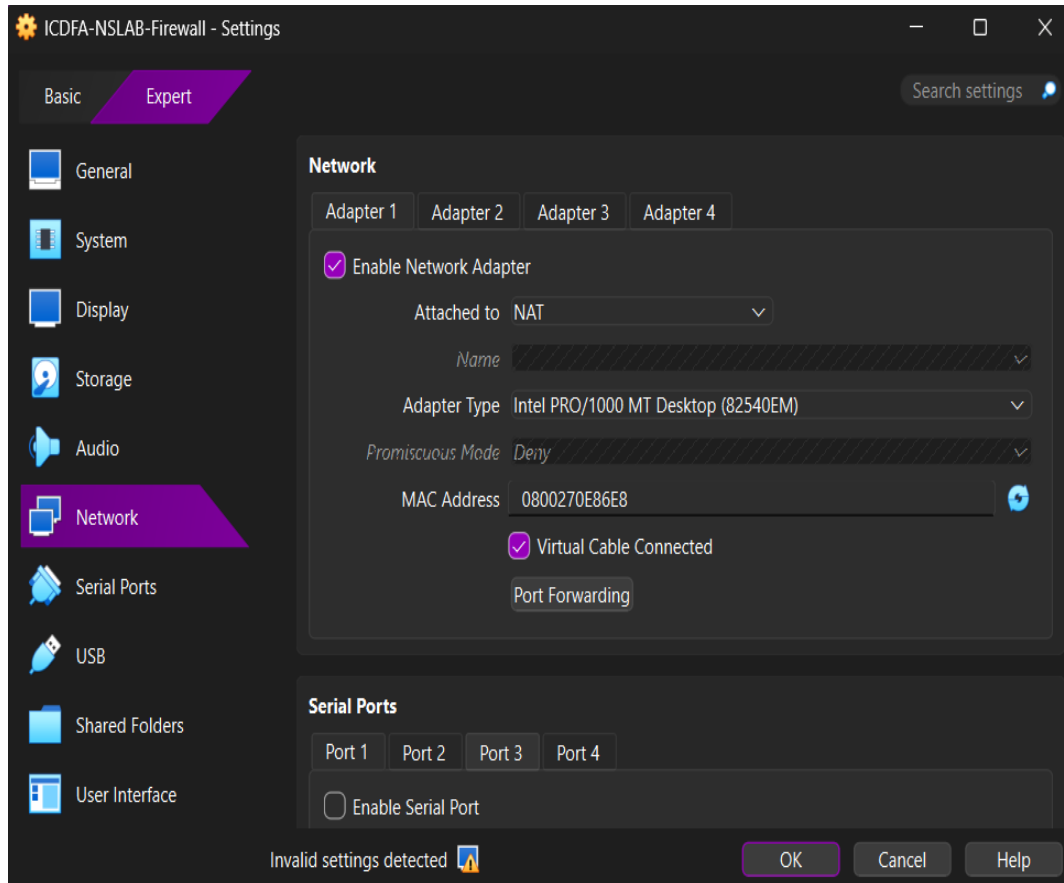
Resource settings

VM	Memory	Processors	Adapters
icdfa-nslab-firewall-v1	2048 MB	2	NAT plus ICDFA-LAN
icdfa-nslab-client-v1	2048 MB minimum	2	ICDFA-LAN only

Part A Preparing VirtualBox

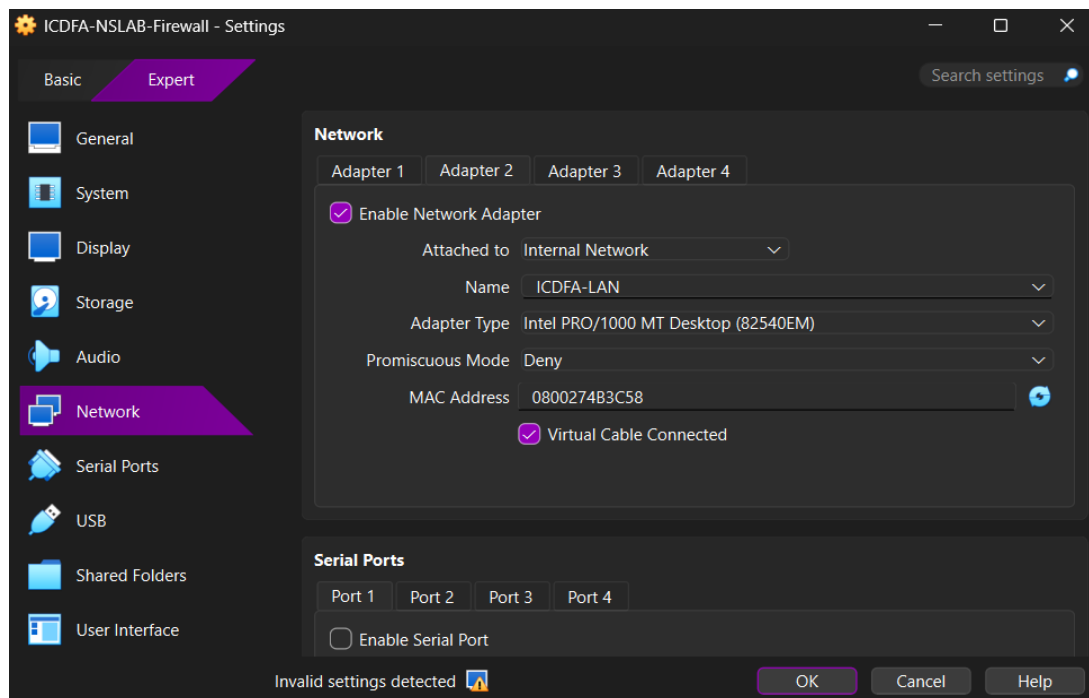
1. Shut down icdfa-nslab-client-v1 and icdfa-nslab-firewall-v1 completely.
2. Selected icdfa-nslab-firewall-v1, then open Settings, and then choose Network.
3. For Adapter 1, select Enable Network Adapter and set Attached to NAT. Confirmed Cable Connected is selected.

Evidence:



- For Adapter 2, selected Enable Network Adapter and Attached to Internal Network. Entered ICDFA-LAN as the name and confirm Cable Connected is selected.

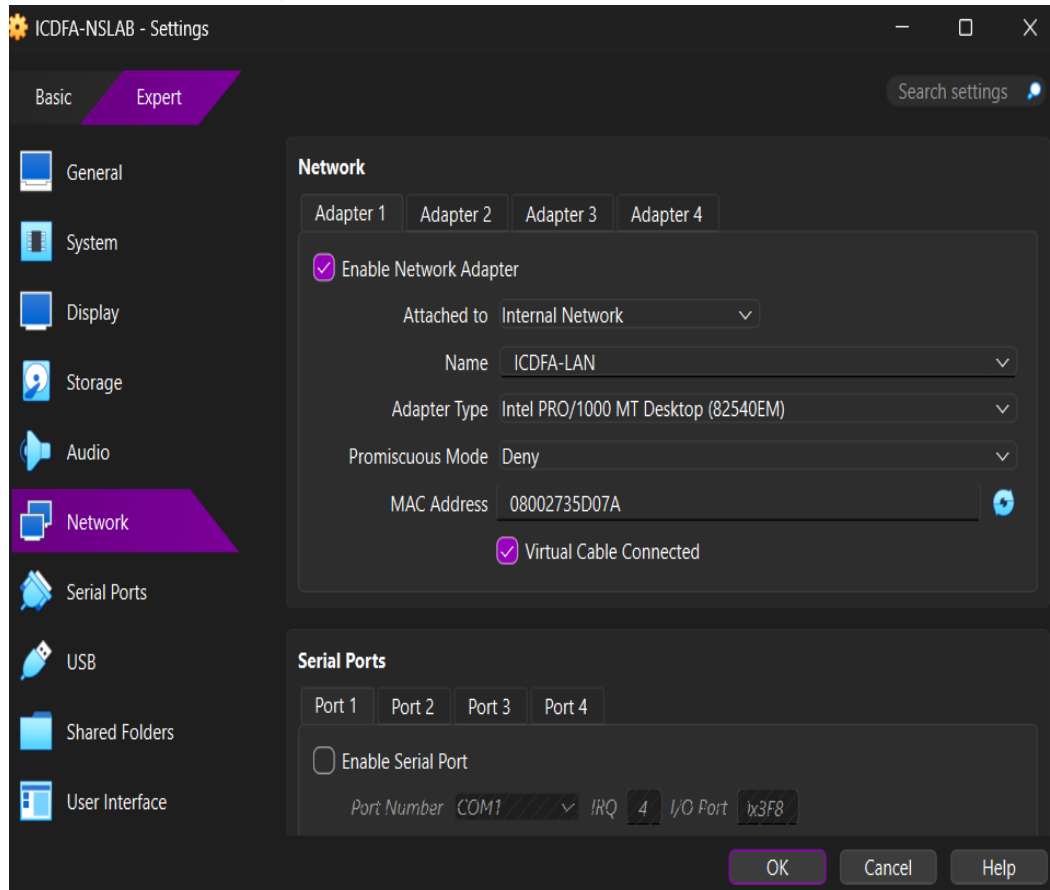
Evidence:



- Disabled any unused Adapter 3 or Adapter 4 for this reduced two-VM laboratory.

CLIENT-NSLAB:

- Select icdfa-nslab-client-v1 and open Settings, then Network.
- For Adapter 1, select Internal Network and enter ICDFA-LAN. Confirm Cable Connected is selected.
- Disabled the remaining Ubuntu adapters..



Part B Verify OPNsense interfaces

- Started the icdfa-nslab-firewall-v1 and waited for the OPNsense console menu to show .
- Confirmed that one interface is marked WAN and the second is marked LAN.
- Confirmed that the LAN address is 10.10.10.1/24.

Evidence:

```

i Code:      https://github.com/opnsense
i Reddit:    https://reddit.com/r/opnsense
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*** icdfa-nslab-firewall-v1.icdfa.test: OPNsense 26.7 (amd64) ***

DMZ (em2)      -> v4: 10.10.20.1/24
LAN (em1)      -> v4: 10.10.10.1/24
WAN (em0)      -> v4: 172.16.100.2/30

HTTPS: SHA256 E8 0B 30 3E D5 89 EB 63 A0 D8 83 2F E6 38 4C 22
          82 2C 5A A4 39 D5 45 54 57 50 A0 73 58 E0 46 90
SSH:   SHA256 K10LtdmW484dUmuPy37tD714RA0De5LKjSU/+bKB1Tw (ECDSA)
SSH:   SHA256 5wvyP8AD885sP0fHz7CgfNStFMhPwXXX42x23cBxco4 (ED25519)
SSH:   SHA256 6e/0XrjSdfL/Hsy8atZw0IPQGsb1P6zFfuJ/FZ50d+A (RSA)

```

Part C Verify Ubuntu addressing

12. Started icdfa-nslab-client-v1 only after icdfa-nslab-firewall-v1 is ready.
13. Open Terminal and display the IPv4 interfaces.

```
ip -4 -br address
ip route
```

Evidence:

```
icdfa@icdfa-nslab-client-v1: ~
icdfa@icdfa-nslab-client-v1:~$ ip -4 br address
Object "br" is unknown, try "ip help".
icdfa@icdfa-nslab-client-v1:~$ ip -4 -br address
lo                UNKNOWN        127.0.0.1/8
enp0s3            UP                10.10.10.102/24
icdfa@icdfa-nslab-client-v1:~$ ip route
default via 10.10.10.1 dev enp0s3 proto dhcp src 10.10.10.102 metric 20100
10.10.10.0/24 dev enp0s3 proto kernel scope link src 10.10.10.102 metric 100
icdfa@icdfa-nslab-client-v1:~$
```

Part D Perform connectivity tests

Run the tests in the sequence below. Do not skip directly to the internet test because each command verifies a different layer.

Test	Command	Expected result	What it proves
1	ping -c 4 10.10.10.1	Replies received	Ubuntu can reach the firewall LAN interface
2	ping -c 4 1.1.1.1	Replies received	Routing and outbound NAT are working
3	getent hosts opnsense.org	One or more addresses returned	DNS name resolution is working
4	curl -I https://opnsense.org	HTTP response headers	TCP, TLS and web connectivity are working

Evidence:

```
icdfa@icdfa-nslab-client-v1:~$ ping -c 4 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data:
64 bytes from 1.1.1.1: icmp_seq=1 ttl=62 time=43.4 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=62 time=84.2 ms
64 bytes from 1.1.1.1: icmp_seq=3 ttl=62 time=34.5 ms
64 bytes from 1.1.1.1: icmp_seq=4 ttl=62 time=344 ms

--- 1.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3003ms
rtt min/avg/max/mdev = 34.506/126.416/343.596/126.777 ms
icdfa@icdfa-nslab-client-v1:~$ getent hosts opnsense.org
2001:1af8:2050:a001:1::1 opnsense.org
icdfa@icdfa-nslab-client-v1:~$ curl -I https://opnsense.org
HTTP/1.1 200 OK
Date: Wed, 30 Sep 2026 04:27:37 GMT
Server: Apache
Link: <https://opnsense.org/wp-json/>; rel="https://api.w.org/", <https://opnsense.org/wp-json/wp/v2/pages/199>; rel="alternate"; title="JSON"; type="application/json", <https://opnsense.org/>; rel=shortlink
Strict-Transport-Security: max-age=15552000; includeSubDomains; preload
Content-Type: text/html; charset=UTF-8
icdfa@icdfa-nslab-client-v1:~$
```

Part E Observe packets in Wireshark

14. Opened e Wireshark on Ubuntu.
15. Selectd the active Ethernet interface carrying the 10.10.10.1 address and begin capture.
16. In Terminal, repeat ping -c 4 10.10.10.1 and getent hosts opnsense.org.
17. Stop the capture after the commands finish.
18. Apply each display filter below and inspect the matching packets.

Filter	What to identify
arp	The request asking who has 10.10.10.1 and the reply containing the firewall MAC address.
icmp	Echo request packets sent by Ubuntu and echo reply packets returned by OPNsense.
dns	The DNS query for opnsense.org and the corresponding response.

Evidences:

*enp0s3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Current filter: arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	PCSSystemtec_35:d0:...	Broadcast	ARP	42	Who has 10.10.10.1?
2	0.004510741	PCSSystemtec_4b:3c:...	PCSSystemtec_35:d0:...	ARP	60	10.10.10.1 is at 08
13	31.560676411	PCSSystemtec_35:d0:...	PCSSystemtec_4b:3c:...	ARP	42	Who has 10.10.10.1?
14	31.568032764	PCSSystemtec_4b:3c:...	PCSSystemtec_35:d0:...	ARP	60	10.10.10.1 is at 08

Frame 1: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface *enp0s3

Ethernet II, Src: PCSSystemtec_35:d0:7a (08:00:06:04:00:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

Address Resolution Protocol (request)

```

0000  ff ff ff ff ff ff 08 00 27 35 d0 7a 08 00 06 04 00 01
0010  08 00 06 04 00 01 08 00 27 35 d0 7a 0a 0a 0a 01
0020  00 00 00 00 00 00 0a 0a 0a 01
  
```

*enp0s3

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

icmp

No.	Time	Source	Destination	Protocol	Length	Info
3	0.004519761	10.10.10.102	10.10.10.1	ICMP	98	Echo (ping) request
4	0.007890086	10.10.10.1	10.10.10.102	ICMP	98	Echo (ping) reply
5	1.006952343	10.10.10.102	10.10.10.1	ICMP	98	Echo (ping) request
6	1.011512663	10.10.10.1	10.10.10.102	ICMP	98	Echo (ping) reply
9	26.116373364	10.10.10.102	10.10.10.1	ICMP	98	Echo (ping) request
10	26.120036865	10.10.10.1	10.10.10.102	ICMP	98	Echo (ping) reply
11	27.118269089	10.10.10.102	10.10.10.1	ICMP	98	Echo (ping) request
12	27.121925699	10.10.10.1	10.10.10.102	ICMP	98	Echo (ping) reply

▶ Frame 3: 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface enp0s3
 ▶ Ethernet II, Src: PCSSystemtec_35:d0:7a (08:00:27:4b:3c:58), Dst: 08:00:27:35:d0:7a (08:00:27:35:d0:7a)
 ▶ Internet Protocol Version 4, Src: 10.10.10.102, Dst: 10.10.10.1
 ▶ Internet Control Message Protocol

0000 08 00 27 4b 3c 58 08 00 27 35 d0 7a 08 00
 0010 00 54 3a c6 40 00 40 01 d7 68 0a 0a 0a 66
 0020 0a 01 08 00 1f 06 0c a8 00 01 42 91 bc 6a
 0030 00 00 09 82 05 00 00 00 00 00 10 11 12 13
 0040 16 17 18 19 1a 1b 1c 1d 1e 1f 20 21 22 23
 0050 26 27 28 29 2a 2b 2c 2d 2e 2f 30 31 32 33
 0060 36 37

Troubleshooting guide given by instructor:

Problem	Likely cause	Corrective action
Ubuntu receives no 10.10.10.x address	DHCP is disabled or the internal-network names do not match	Check both adapters use ICDFA-LAN; enable DHCP on OPNsense LAN; renew the Ubuntu lease.
10.10.10.1 cannot be pinged	LAN adapter is disconnected or assigned incorrectly	Confirm Cable Connected, verify interface assignment, and check the OPNsense LAN address.
1.1.1.1 fails but 10.10.10.1 works	WAN, default route or outbound NAT problem	Confirm OPNsense WAN uses DHCP on NAT and review System Routes Status.
Names fail but 1.1.1.1 works	DNS problem only	Review DHCP DNS settings and Services Unbound DNS; renew the Ubuntu lease.
Web interface warning appears	The firewall uses a self-signed certificate	Proceed only to https://10.10.10.1 inside the authorised lab; do not ignore such warnings on unknown public sites.